Agcam

WALTER SCOTT, JR.
COLLEGE OF ENGINEERING
COLORADO STATE UNIVERSITY

Plant Finder Design Challenge Instructions

Abstract

Each student on the Greenhouse team majoring in an ECE discipline is challenged to make a plant finder method or methods for the design challenge, due on Wednesday 28 October 2026 at the team meeting (4:00 pm). Submissions will be evaluated in person based on the files in your pull request(s).

Note 1: please run *git pull origin main* while in your branch to get new changes to the codebase, as of today 20 Oct 2026.

Note 2: The evaluation is subject to change if noticeable errors are seen from a pixel by pixel evaluation over the test set, however, all detectors will be evaluated on the same metric. All analysis will be respective to the design implementation, and false positives are to be penalized heavier in scoring than false negatives.

1 Design Challenge Instructions

For this design challenge, you are tasked with creating an entirely new plant detection method for the Agcam automated greenhouse that has a greater accuracy than the legacyHSV method. Submit an entry from a branch you already opened named with your first name(s) by running:

```
git add plant_finders/descriptiveWords_plant_finder.py
git commit -m "submission"
git push origin HEAD
```

Then open a pull request on Github and assign Kai or Abby. The pull request is subject to the deadline, pushing alone is not enough.

Helpful hints:

1. You may reference the legacyHSV method to see a way that the color based plant finder was previously coded. The current legacyHSV has an accuracy of about 60% (pixel wise, as defined in the test_accuracy method in the accuracy test) on the current image.
2. There are many types of detection with openCV, i.e. color based, edge based, etc. There are also other ways to detect plants, like AI models, etc. Creativity may help in this challenge.
3. More test files will become available throughout the challenge, be sure to *git pull origin main* every once in a while to profit from them.

4. Locally installing more libraries is permitted.

5. Working together is permitted, and both students will be evaluated equally for a shared submission. AI use is expected to be in line with course and university policy.

6. You are not limited to one submission. You may create multiple plantfinders, even with small differences.

2 Style Guide

File Naming: Each plant detection module must be named and placed as follows:
plant_finders/descriptiveWords_plant_finder.py

2.1 Main Method

1. **Method Naming:** find_plants
   Each module is expected to have the main method find_plants. You may have other helper methods.

2. **Inputs:** image_file_path
   As inputs to the main module, you may take in the file path to the image. Do not take in any other parameters, as the modules should be able to be switched out 1:1. (However, you may use optional arguments if you want)

3. **Returns:** mask
   Returns a mask of the detected plant matter in the same dimensions as the original image. To be precise (quoting from https://pyimagesearch.com/2014/08/04/opencv-python-color-detection/)

   > a binary mask is returned, where white pixels (255) represent pixels that fall into the upper and lower limit range and black pixels (0) do not.

   You do not need to use openCV to get this output, but it must be formatted this way.

eg. plant_finders/legacyHSV_plant_finder.py main method: find_plants(image_file_path) which returns a mask.

2.2 Style Guide

1. Any stages in a pipeline that are necessary for the module should be implemented as functions for modularity and readability.

2. The code should be organized with the main function first and helper functions after.

3. Functions that are used in multiple modules should be duplicated in each module to make deletion easier, unless they are so general to openCv, image processing, etc that they belong in a respective util.

2.3 Comments

As it is a principal part of the code, every plant detection module must include the following header comment:

```
"""
Copyright         : <Year> <Copyright Owner>
File Name         : <Filename>
Description       : <Write your file description here>

Revision History  :
Date  Author    Comments
--------------------------------------------------------------------
07/11/2014 <Person working on the file> <Short change description>
"""
```

There should also be internal comments for each function in the following format, within and immediately following the def:

```
    """
<function description>

Args:
    <name>: <description>

Returns:
    <description>
"""
```

3 How to Test Your Code

You may test your code with the files provided in *tests/plant_finder_functionality_test.py* and *tests/plant_finder_accuracy_test.py*.

To test your method, just update the import for plant_finder to be your plantfinder file instead of legacyHSV. You may also change the test and key images in the method calls. **Please do not commit changes to the test files.** To run the test, run *python3 tests/plant_finder_accuracy_test.py* (or python if needed for your venv)